Session 6: Python Fundamentals

Detailed Theory

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1. Installing Python

- **Python** is a popular, high-level programming language known for readability and ease of use.
- To install Python, go to python.org, download the latest version for your OS (Windows, Linux, macOS).
- Installation steps usually involve running the installer and ensuring "Add Python to PATH" option is selected for command-line access.
- To verify installation, open terminal/command prompt and run:

  bash

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python --version

- Alternatively, python3 --version is used on some systems.

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2. Your First Python Program

- Python programs typically have .py extension.
- To write a simple program, create a file hello.py with:

python

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print("Hello, World!")

- Run it via terminal with:

bash

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python hello.py

- print() is a built-in function to output text.

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3. Declaring Functions

- Functions are blocks of reusable code defined using the def keyword.

- Syntax:

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  def function_name(parameters): # code block return value

- Example:

  python

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```
def greet(name): print("Hello, " + name + "!")
```

- Functions help organize code and avoid repetition.

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4. What’s an Object?

- Python is **object-oriented**. Everything in Python is an object.
- An **object** is an instance of a class, representing data and behavior.
- Example: Strings, numbers, lists, functions — all are objects.
- You can create your own classes to define custom objects.

5. Indenting Code

- Python uses **indentation** (whitespace at the start of a line) to define code blocks.
- Indentation replaces braces {} used in many other languages.
- Proper indentation is critical; incorrect indentation causes IndentationError.
- Example:

  python

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  ```
  if 5 > 2: print("Five is greater than two")
  ```

- The line after if is indented to indicate it belongs to the block.

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6. Testing Modules

- Python code can be organized in **modules** — .py files containing functions and classes.
- You can import modules using import keyword:

python

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import math print(math.sqrt(16))

- To test modules, the if __name__ == "__main__": block is used to run test code only when the module is executed directly, not when imported:

  python

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  def main(): print("Testing module") if __name__ == "__main__": main()

- This is useful for debugging and modular development.

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7. Native Data Types

- Python supports various built-in/native data types:

Data Type	Description	Example
int	Integer numbers	5, -10, 0
float	Floating-point numbers (decimals)	3.14, -0.001
str	Text strings	"Hello", 'Python'
bool	Boolean values	True, False
list	Ordered, mutable collection	[1, 2, 3]
tuple	Ordered, immutable collection	(1, 2, 3)
dict	Key-value pairs	{'name': 'Komal', 'age': 25}
set	Unordered collection of unique elements	{1, 2, 3}
NoneType	Represents the absence of value	None

- These types form the foundation for Python programming.